

# Ahmed Muharram

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## SUMMARY

Applied AI/ML engineer building production **tool-calling LLM agents**, **Retrieval-Augmented Generation (RAG)** systems, and high-throughput backend infrastructure end-to-end. UPenn MSE/BSE in Computer Science; Master's thesis defended May 2026, contributing two open-source scientific LLM agents on a pydantic-ai / LangGraph / pgvector stack including **PathFinder**, **which is in active integration with the VEuPathDB pathogen-genomics consortium** (14 databases used by infectious-disease labs worldwide). At Fidelity Investments, built a hybrid PostgreSQL FTS + pgvector retrieval system to over 200 daily R&D users with terabyte-scale ingestion at 99.99%+ success.

**Egyptian citizen. Military service: Exempt.**

## EDUCATION

**University of Pennsylvania** School of Engineering and Applied Science Philadelphia, PA

Master of Science in Engineering (M.S.E.) in Computer Science, GPA 3.82 Apr 2024 – May 2026

Bachelor of Science in Engineering (B.S.E.) in Computer Science, GPA 3.75 Aug 2022 – May 2026

**Onsi Sawiris Scholar** fully-funded merit-based scholarship awarded to top Egyptian students for study at leading US universities (UPenn, Harvard, MIT, Stanford, UChicago)

*Minors:* Mathematics, Data Science, and Engineering Entrepreneurship

## EXPERIENCE

**AI/ML Engineer**, Fidelity Investments Dec 2025 - Jan 2026

- Productionized and scaled an internal LLM/RAG system from intern phase to production; expanded support to arbitrary enterprise data sources via a standardized connector / ingestion / indexing template, enabling rapid onboarding of new corpora.
- Replaced the BM25S retrieval index with a custom **hybrid PostgreSQL Full-Text Search + pgvector** design, cutting **index refresh latency from 10 minutes per 1M chunks to under 1 second** while maintaining retrieval quality under continuous corpus updates.
- Implemented **multivector / late-interaction retrieval** with **hybrid-score reranking**, sharded the Qdrant cluster, and backfilled historical embeddings into pgvector to unify legacy and new corpora. System now used in production by the R&D teams (~200 daily users).

**Data Science and AI Intern**, Fidelity Investments Jun 2025 - Aug 2025

- Architected a high-throughput data ingestion pipeline processing **terabytes of internal data** with **99.99%+ ingestion success**, using schema validation, retries, and idempotent writes for reliability under production load.
- Built an LLM/RAG application on **Qdrant + pgvector** grounding answers in enterprise documents; added hallucination/grounding checks and an evaluation harness tracking retrieval accuracy and failure modes.
- Containerized and deployed services with **Docker and Kubernetes** for reproducible environments and clean handoff to downstream teams.

**Software Engineering Intern (AI Integration)**, Aydi Jun 2024 - Aug 2024

- Built a low-connectivity-friendly chatbot for agricultural business owners on Haystack, MongoDB Atlas, Redis, and GPT-4o, achieving **85%+ top-1 retrieval accuracy** on held-out farm analytics queries.
- Added token/latency observability and Jest test coverage; **reduced LLM spend by 96.25%** via a cost/quality-driven model migration from GPT-4o to GPT-4o mini.

## PROJECTS

**PathFinder (Pathogen Finder): Production Tool-Calling LLM Agent for the VEuPathDB Consortium**

- Built a **44-tool LLM agent** that turns natural-language biological questions into executable multi-step **strategy graphs** (recursive ASTs of WDK searches, Boolean operators, and ortholog transforms) over the **14-database VEuPathDB consortium** (PlasmoDB, ToxoDB, CryptoDB, FungiDB, VectorBase, TriTrypDB, GiardiaDB, MicrosporidiaDB, and others). **Currently being integrated into the VEuPathDB platform.**
- Backend: **pydantic-ai** agents on a **LangGraph** multi-phase pipeline (scoping, discovery, planning, execution, verification) with AsyncPostgresSaver checkpointing for durable, resumable chat threads; Vercel AI SDK v6 **Server-Sent**

**Events (SSE)**, semantic catalog search via **nomic-embed-text-v1.5** + **pgvector HNSW**, cross-thread memory store, and durable long-running tools on **Procrastinate** workers with Tenacity-based retry/backoff.

- Observability: full **OpenTelemetry** instrumentation (FastAPI, SQLAlchemy, httpx) exporting to **SigNoz**; LLM tracing via **Langfuse**. Frontend on Next.js 16 / React 19 / TanStack Query / Zustand / xstate state machines with a **ReactFlow** strategy graph editor; Playwright + axe-core E2E suite.

## ScenIQ: Enterprise AI Meeting Intelligence

*UPenn Senior Project*

- Built an **on-prem, security-first** meeting intelligence platform turning meeting video into transcripts, topics, and structured action items via a multi-stage pipeline: FFmpeg, speaker diarization, Whisper transcription, segmentation, and action-item extraction. **Runtime: ~1-2 minutes per 5-minute video.**
- FastAPI + PostgreSQL + pgvector backend with speaker / org-chart embeddings for relationship-aware action assignment; Zoom/Teams ingestion, ClickUp/Jira outputs; React + TypeScript + Vite frontend; air-gapped / private-cloud deployment with reproducible eval tooling.

## ByteNet: Distributed Web Search Engine · PennOS: UNIX-like OS Simulator

- **ByteNet**: end-to-end distributed search engine on a custom Java stack (webserver + KV store + RDD-style framework) across an AWS EC2 coordinator/worker cluster; production-grade web crawler at **~500k URLs** with **robots.txt** compliance and content-hash deduplication; distributed TF-IDF inverted index + iterative PageRank served via a **/search JSON API**.
- **PennOS**: UNIX-style process model on **s pthreads** with PCB design and split user/kernel syscall layers; SIGALRM preemptive scheduler with 100ms quanta and 3-level priority queues; **PennFAT (FAT16-style)** filesystem inside a host file using a memory-mapped FAT region with full **open/read/write/seek/unlink** semantics.

## RESEARCH

### Master's Thesis (defended May 2026): How Underspecified Prompts Shape Tool-Calling LLM Agents in Scientific Workflows

*Advisor: Chris Callison-Burch | Reader: Eric Wong | Collaborators: David S. Roos (VEuPathDB), Delip Rao, Andrew Zhu*

- Two case studies of tool-calling LLM agents in real scientific toolchains: materials feasibility (LAMINA) and pathogen genomics (PathFinder, see Projects). Cross-domain finding across 5 OpenAI model configurations: **prompt under-specification dominates model scale, routing policy, and task complexity** as a source of agent failure; instability localizes to the natural-language-to-tool-invocation mapping, not to tool execution.
- **LAMINA** (50-tool materials-feasibility agent over CALPHAD, Materials Project DFT, and CHGNet neural surrogates): evaluated on **DARPA SciFy Drop-4** (37 ordinal feasibility claims, 3 OpenAI models incl. gpt-5.1) using **Quadratic Weighted Kappa**; vague claims induce a **flip rate of 0.43**.
- **PathFinder evaluation: 420-run controlled study** over a 35-strategy gold corpus (8 VEuPathDB sites, 31 WDK search types) using gene-set F1, search accuracy, Parameter Fidelity, and Zhang-Shasha tree edit distance with 95% bootstrap BCa confidence intervals; gene-set F1 **collapses 0.66 / 0.23 / 0.16** across precise / moderate / vague prompts. Derived four production design recommendations: disambiguation protocols, catalog grounding, incremental validation, replicated evaluation.

## SKILLS

**Tool-Calling Agents & LLM Frameworks:** LLMs, Tool-Calling / Function-Calling Agents, Multi-Agent Systems, Agentic AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Routing, Guardrails; pydantic-ai, LangGraph, LangChain, Haystack, Kani, Vercel AI SDK; OpenAI / Anthropic / Gemini APIs

**Retrieval, Embeddings & LLM Ops:** pgvector (HNSW), Qdrant, BM25, hybrid / multivector / late-interaction retrieval, hybrid-score reranking, sentence-transformers (nomic-embed), fastembed, Whisper; OpenTelemetry, Langfuse, SigNoz; LLM evaluation (QWK, F1, bootstrap CIs, ICC)

**Backend & Languages:** Python, TypeScript, JavaScript, SQL, Java, C; FastAPI, Pydantic, SQLAlchemy, asyncpg, Alembic, Procrastinate; REST, OpenAPI, WebSockets, Server-Sent Events (SSE); microservices, distributed systems, CQRS

**Frontend:** React, Next.js, Vite, Tailwind, Zustand, TanStack Query / Table, ReactFlow, Zod, assistant-ui

**Cloud, Data & DevOps:** PostgreSQL, Redis, MongoDB; Docker, Kubernetes, AWS (EC2, RDS), Linux, Bash, Git, GitHub Actions, CI/CD; Pytest, Vitest, Jest, Playwright, axe-core, Stryker (mutation testing)

**Spoken:** Native Arabic and English; limited German and Italian

## AWARDS

**NASA Space Apps Cairo Special Mentorship Award**, Nilepreneurs (out of 115 teams and 859 participants)